DAY 03 STUDY NOTES • PHASE 0: HTML & CSS FOUNDATIONS

Day 3 — CSS Fundamentals, Box Model, Positioning & Flexbox

React + Next.js Roadmap | Complete Beginner Learning Notes with Examples, Visual Outputs, and Practice

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Stack Track: Full-Stack Java to React / Next.js
Topic Focus: Presentation, Layouts & Dimensional Flow

Executive Study Blueprint & Pedagogical Objective

This study guide documents the core visual architecture of frontend engineering mastered during **Day 3**. Moving beyond raw HTML markup, we explore how **Cascading Style Sheets (CSS)** transform structural elements into beautiful, responsive interfaces. This manual covers CSS syntax, selector specificity, the mathematical geometry of the CSS Box Model, typography metrics, color spaces, the generic `<div>` container, inline vs block display types, spatial positioning schemes (relative, absolute, fixed, sticky), and single-dimensional layout architecture using **Flexbox**.

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Chapter 1: CSS Introduction

CSS stands for **Cascading Style Sheets**. While HTML provides the raw semantic structure (the skeleton), CSS dictates the visual presentation, aesthetic styling, typography, colors, and layout positioning (the skin and clothes).

Anatomy of a CSS Rule

A CSS rule consists of a **selector** pointing to the HTML element to style, followed by a **declaration block** enclosed in curly braces `{}`. Inside the declaration block, one or more declarations define a **property** and its corresponding **value** separated by a colon, terminating with a semicolon.

```
h1 { color: purple; font-size: 32px; } ^ ^ ^ ^ ^ | | | | +-- Value (assigned style) | | +-- Pr
```

Source HTML

```
<h1>Hello CSS</h1>
<p>I am learning frontend development.</p>
```

Target CSS

```
h1 {
  color: purple;
  font-size: 32px;
}

p {
  color: gray;
  font-size: 20px;
}
```

EXPECTED BROWSER VISUAL OUTPUT:

Hello CSS

I am learning frontend development.

The Three Ways to Apply CSS

- **1. Inline CSS:** Written directly within an element's opening tag using the `style` attribute (e.g., `<h1 style="color: purple;">`). *Disadvantage:* Violates separation of concerns and cannot be reused across pages.
- **2. Internal CSS:** Placed inside a `<style>` element nested within the HTML document's `<head>`. *Disadvantage:* Applies only to a single page.
- **3. External CSS (Industry Standard):** Stored in a standalone `.css` file and linked to any number of HTML documents via a `<link>` tag in the `<head>`. Enables site-wide styling and browser caching.

Linking an External Stylesheet

```
<link rel="stylesheet" href="style.css">
```

- `<link>`: An empty void tag that establishes a relationship between the HTML document and an external resource.
- `rel="stylesheet"`: Specifies the *relationship*. Tells the browser that the linked file contains styling rules to parse and apply to the DOM.
- `href="style.css"`: The *hypertext reference* indicating the relative or absolute file path to the stylesheet.

Chapter 2: CSS Selectors

CSS selectors define which DOM nodes are targeted for styling. Understanding selector syntax and specificity prevents unintended visual regressions.

HTML Structure

```
<h1 id="main">My Website</h1>

<p class="description">First paragraph</p>

<p class="description">Second paragraph</p>

<button class="btn">Click Me</button>
```

Associated CSS Rules

```
/* 1. Element Selector */
h1 {
  color: purple;
}

/* 2. Class Selector */
.description {
  color: gray;
}

/* 3. ID Selector */
#main {
  font-size: 32px;
}

/* 2b. Reusable Class */
.btn {
  background-color: blue;
  color: white;
}
```

Selector Type	Prefix Symbol	Targeting Behavior	Engineering Rule & Reusability
Element Selector	None (Tag name)	Targets all matching HTML tags globally (e.g. `h1`, `p`, `button`).	Used for global resets and baseline typography. Low specificity.
Class Selector	Period / Dot (`.`)	Targets any element carrying the matching class name (`.description`).	**Highly reusable:** Many elements can share the same class across the entire document.
ID Selector	Hash / Pound (`#`)	Targets the single element possessing that unique ID (`#main`).	**Strictly unique:** An ID must appear only once per HTML document. High specificity

Why Classes Use Dots and IDs Use Hashes: The dot (`.`) signals to the CSS parser that the following token represents a CSS class attribute, while the hash (`#`) maps directly to the DOM `id` attribute. In modern UI development (including React and Next.js), class selectors (`.btn` , `.card`) are the preferred standard for reusability, while IDs are reserved for form bindings and URL fragment anchors.

Chapter 3: The CSS Box Model

In CSS, every single element rendered on screen is treated as a rectangular box. The **CSS Box Model** governs how the geometry of this rectangular box is calculated from the inside out:

```
.card {
  width: 300px;
  padding: 20px;
  border: 2px solid black;
  margin: 30px;
}
```

The Concentric Layers of the Box Model

```
        +-------------------------------------------------------+ | MARGIN: 30px (Transparent space o
```

Total Rendered Width Calculation (Standard Content-Box)

Under the default CSS standard (`box-sizing: content-box`), declared `width` applies *only to the innermost content area*. Padding and borders add to the total rendered footprint:

`Total Rendered Width = Content (300px) + Left Padding (20px) + Right Padding (20px) + Left Border (2px) + Right Border (2px) = 344px`

Note on Margin: Margin (30px on each side = 60px horizontal space) pushes neighboring elements away. It adds to the element's overall layout footprint on the page, but sits outside the element's border and does not change the element's internal box size.

Chapter 4: Margin vs Padding & Shorthand Syntax

While both properties generate blank breathing room, they operate on opposite sides of the border:

Padding (Inside the Border)

- Creates space *between the content and the border*.
- Adopts the element's `background-color` .
- Expands the clickable hit-area of buttons and links.

Margin (Outside the Border)

- Creates separation *between the border and neighboring elements*.
- Always completely transparent (reveals the parent background).
- Can collapse vertically with adjacent sibling margins.

The Clockwise Shorthand System (TRBL)

```
        Four-value shorthand: padding: 10px 20px 30px 40px; | | | | Top -------+ | | | Right ---------
```

Shorthand Pattern	Code Example	Expanded Side-by-Side Equivalents
1 Value (All Sides)	`padding: 10px;`	Top, Right, Bottom, and Left are all set to **10px**.
2 Values (Vertical / Horizontal)	`margin: 15px 25px;`	**15px** Top & Bottom; **25px** Left & Right.
3 Values (Top / Horiz / Bottom)	`padding: 10px 20px 30px;`	**10px** Top; **20px** Left & Right; **30px** Bottom.
4 Values (Clockwise TRBL)	`padding: 10px 20px 30px 40px;`	**10px** Top; **20px** Right; **30px** Bottom; **40px** Left.

Chapter 5: CSS Typography

Typography controls the legibility, hierarchy, and reading rhythm of text. CSS provides fine-grained controls over typefaces, sizing, weights, line spacing, and alignment:

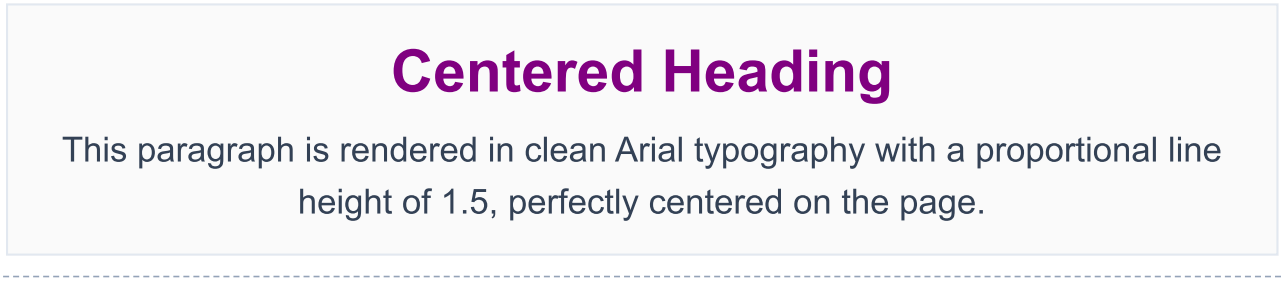
```
body {
  font-family: Arial, sans-serif;
}

h1 {
  font-size: 32px;
  color: purple;
  text-align: center;
}

p {
  font-size: 20px;
  font-weight: 400;
  line-height: 1.5;
  text-align: center;
}
```

- `font-family: Arial, sans-serif;` : Defines the prioritized font stack. The browser tries to load Arial; if unavailable on the user's operating system, it falls back to a generic `sans-serif` system font.
- `font-size` : Sets the physical height of characters (e.g. 32px for h1, 20px for body text).
- `font-weight: 400;` : Dictates stroke thickness. 400 corresponds to normal body weight; 700 corresponds to bold.
- `line-height: 1.5;` : Unitless proportional multiplier setting vertical line spacing (1.5 times the font size), preventing cramped, unreadable text lines.
- `text-align: center;` : Aligns text horizontally relative to its block container (`left` , `center` , `right` , or `justify`).

EXPECTED BROWSER VISUAL OUTPUT:



Chapter 6: CSS Colors and Backgrounds

In CSS, foreground typography color is declared via the `color` property, whereas container surface tint is declared via `background-color` .

```css
h1 {
  color: purple;                        /* 1. Named Color Keyword */
}

body {
  background-color: #f5f5f5;            /* 2. Hexadecimal (HEX) Code */
}

p {
  color: #333333;                       /* 2b. Dark Slate HEX Code */
}

button {
  background-color: rgb(37, 99, 235);   /* 3. RGB Functional Notation (Red, Green, Blue) */
  color: white;
}
```

Color Notations Explained

- **Named Colors:** Standardized keywords like `purple`, `lightblue`, `white`.
- **HEX Colors:** 6-character hex strings representing red, green, and blue intensities from `00` to `FF` (e.g. `#f5f5f5`).
- **RGB Colors:** Functional notation accepting integer values from 0 to 255: `rgb(red, green, blue)`.

The "Light Blue" Syntax Trap

Incorrect: `background-color: light blue;` CSS treats spaces as property value separators. The browser tries to parse "light" and "blue" as two conflicting values and completely ignores the rule.

Correct: `background-color: lightblue;` (a single keyword token).

Day 3 Practiced Color Palette

purple	lightblue	rgb(37, 99, 235)
Text Heading	Card Background	Button Accent

Chapter 7: CSS Borders and Border Radius

Borders define visible boundaries around elements, and `border-radius` softens those boundaries into rounded corners or circular avatars:

```css
.card {
  border: 2px solid black;
  border-radius: 10px;
}

button {
  border: none;
  border-radius: 5px;
}
```

Deconstruction of the Border Shorthand: `border: 2px solid black;`

- **2px:** `border-width` (thickness of the stroke).
- **solid:** `border-style` (line pattern: `solid`, `dashed`, `dotted`, or `none`).
- **black:** `border-color` (the color tint of the stroke).
- `border: none;`: Removes the default beveled gray outline that browsers automatically render on `<button>` elements.

Border Radius Corner Geometry Shapes

Square Corners (`border-radius: 0px`)
Sharp, 90-degree angular edges.

Slightly Rounded (`border-radius: 5px`)
Subtle modern curve for buttons and inputs.

Rounded Card (`border-radius: 14px`)
Smooth container card aesthetic.

Pill / Circular Shape (`border-radius: 50%` / pill)
Fully rounded oval capsule or circle.

Chapter 8: The HTML Element

The `<div>` tag stands for **division**. It is a completely generic, non-semantic block-level container element used exclusively to group related child elements together for CSS styling and layout manipulation.

**Grouping Elements Inside a **

```
<div class="card">
  <h2>Frontend Developer</h2>
  <p>I am learning CSS.</p>
  <button>Learn More</button>
</div>
```

Styling the Container via Class

```
.card {
  width: 300px;
  background-color: lightblue;
  padding: 20px;
  border: 2px solid black;
  border-radius: 10px;
}
```

Parent-Child Hierarchy Tree

```
        div.card (Parent Container: encapsulates dimensions, background & borders) ├── h2 (Child Node
```

> **Semantic Elements vs :** Use semantic elements (`<header>` , `<main>` , `<section>` , `<article>` , `<footer>`) whenever content has a clear semantic role on the page. Use `<div>` purely as an unopinionated styling wrapper or layout scaffolding when no semantic tag naturally applies.

Chapter 9: The CSS Display Property

The `display` property dictates how an element behaves in the normal document flow and how it interacts with neighboring elements:

Display Value	Starts on New Line?	Width & Height Supported?	Default Example HTML Elements
`block`	**Yes** (forces full width)	**Fully Supported:** Explicit dimensions are respected.	`<div>` , `<p>` , `<h1>` – `<h6>` , `<section>`
`inline`	**No** (flows with text)	**Ignored:** `width` & `height` have no effect.	`<span>` , `<a>` , `<strong>` , `<em>`
`inline-block`	**No** (flows side-by-side)	**Fully Supported:** Respects width, height, and vertical padding.	`<button>` , `<img>` , `<input>`

1. Block Behavior

```
.box {
  display: block;
  /* Occupies 100% width;
     forces next element
     onto a new line */
}
```

2. Inline Behavior

```
span {
  display: inline;
  /* Wraps around text;
     width and height
     are ignored */
}
```

3. Inline-Block Hybrid

```
.box {
  display: inline-block;
  width: 100px;
  height: 50px;
}
```

> **Why `inline-block` is Essential:** Standard `inline` elements flow alongside text but ignore dimensional properties like `width` , `height` , and vertical margins. Setting `display: inline-block;` gives an element the best of both worlds: it can sit horizontally side-by-side with other elements while fully honoring custom width, height, borders, and margins.

Chapter 10: The Five CSS Positioning Schemes

The `position` property overrides default document flow, allowing elements to be precisely anchored or floated using offset properties (`top` , `right` , `bottom` , `left`):

Position Scheme	In Normal Flow?	Reference Origin Anchor	Typical Practical Use Case
1. `static`	Yes (Default)	Normal document flow. Offsets have no effect.	Standard paragraphs, headings, default layout.
2. `relative`	Yes (Original space reserved)	Its own natural un-positioned starting point.	Slight offsets; anchor parent for absolute children.
3. `absolute`	**No** (Completely removed)	Nearest *positioned ancestor* (non-static parent).	Card notification badges, close modal icons, tooltips.
4. `fixed`	**No** (Removed from flow)	The browser **viewport window**.	Persistent floating "Back to Top" or chat support buttons.
5. `sticky`	Hybrid (Normal flow until scroll)	Scroll container offset threshold.	Sticky navigation bars that pin to top on page scroll.

Relative Positioning (Space Reserved)

```
.box {
  position: relative;
  top: 20px;
  left: 30px;
}
```

The box shifts 20px down and 30px right, but its original layout space remains occupied in the document.

Absolute Badge Anchored to Relative Parent

```
.card {
  position: relative; /* Containing Block */
}

.badge {
  position: absolute;
  top: 10px;
  right: 10px;
}
```

The badge anchors exactly 10px from the top-right corner of the parent card.

Fixed Button Pinned to Viewport

```
button.floating-cta {
  position: fixed;
  bottom: 20px;
  right: 20px;
}
```

Stays permanently pinned to the bottom-right of the screen regardless of how far the user scrolls.

Sticky Header Pinned to Top

```
header.site-header {
  position: sticky;
  top: 0;
}
```

Scrolls naturally until it touches the top edge of the viewport, where it locks in place.

Chapter 11: Flexbox Fundamentals

The **Flexible Box Layout (Flexbox)** module is a 1-dimensional layout system designed to align, distribute, and space items cleanly across a container, even when their dimensions are dynamic:

```
<!-- HTML Structure -->
<div class="container">
  <div class="box">Java</div>
  <div class="box">HTML</div>
  <div class="box">CSS</div>
</div>
```

```css
/* Flexbox Container & Items CSS */
.container {
  display: flex;
  flex-direction: row;
  justify-content: space-between;
  align-items: center;
  height: 200px;
  background-color: white;
}

.box {
  width: 100px;
  height: 50px;
  background-color: lightblue;
  border: 2px solid black;
  text-align: center;
}
```

Understanding the Two Axes (for `flex-direction: row`)

```
        MAIN AXIS (Horizontal: controlled by 'justify-content') ------------------------------------
```

Key Flexbox Properties:

- **`display: flex;`**: Activates the Flexbox layout engine on the parent container, converting its direct children into flex items.
- **`flex-direction`**: Defines the orientation of the main axis (`row`: horizontal left-to-right; `column`: vertical top-to-bottom).
- **`justify-content`**: Governs alignment along the **Main Axis** (`flex-start`, `center`, `flex-end`, `space-between`, `space-around`, `space-evenly`).
- **`align-items`**: Governs alignment along the **Cross Axis** (`stretch`, `flex-start`, `center`, `flex-end`).

> **The Quick Alignment Rule:** In a standard row-based layout, `justify-content` controls horizontal placement across the row, while `align-items` controls vertical placement within the row height.

Chapter 12: Flexbox Gap and Wrapping

By default, flex containers force all items onto a single line (`flex-wrap: nowrap`). The `flex-wrap: wrap;` rule permits overflowing items to break naturally onto subsequent lines:

```css
body {
  background-color: #f5f5f5;
}

.container {
  display: flex;
  flex-wrap: wrap;
  gap: 20px;
  width: 350px;
  height: 250px;
  background-color: white;
  align-content: center;
}

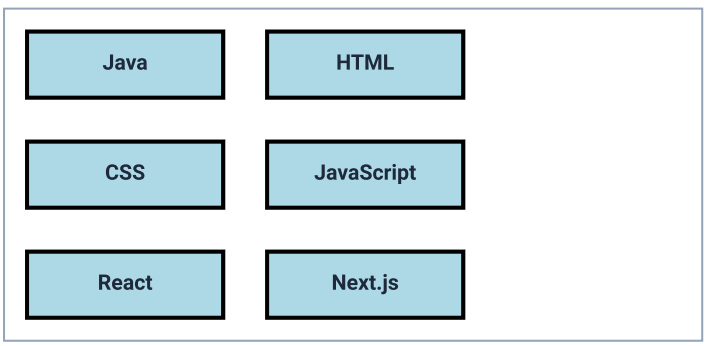
.box {
  width: 100px;
  height: 50px;
  background-color: lightblue;
  border: 2px solid black;
  text-align: center;
}
```

Mathematical Proof of Wrapping (Why the Third Box Drops Down)

1. Each box has a declared content width of 100px plus a 2px border on each side (2px left + 2px right = 4px). Thus, each box has an outer width of **104px**.
2. If three boxes were to sit on a single line, they would require: 104px + 104px + 104px = **312px**.
3. Two intervening gaps of 20px each add: 20px + 20px = **40px**.
4. Total horizontal width required for three boxes: 312px + 40px = **352px**.
5. Because **352px exceeds the container's fixed width of 350px**, the third box cannot fit on the first line. The browser wraps it down to a new line, placing exactly 2 boxes per row across 3 distinct rows.

Property	Scope & Purpose	When Does It Take Effect?
`gap`	Creates empty gutter space between adjacent flex items.	Always active. Eliminates the need for manual margin hacks on items.
`align-items`	Aligns individual items along the cross axis *within their own line*.	Applies to every individual line in both single and multi-line flex containers.
`align-content`	Distributes *entire flex lines* relative to each other along the cross axis.	**Only works when `flex-wrap: wrap;` is active** and there is extra vertical space.

VISUAL SIMULATION: 6 BOXES WRAPPING IN A 350PX CONTAINER



Chapter 13: Complete Practical Project: My Frontend Skills Card

The following project ties together all the Day 3 concepts: typography, background palettes, borders, card grouping, flexbox layout, and gap distribution:

1. Project HTML (`index.html`)

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-
width, initial-scale=1.0">
  <title>My Frontend Skills</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <h1>My Frontend Skills</h1>

  <div class="container">

    <div class="card">
      <h2>Java</h2>
      <p>Backend programming</p>
      <button>Learn More</button>
    </div>

    <div class="card">
      <h2>HTML</h2>
      <p>Web page structure</p>
      <button>Learn More</button>
    </div>

    <div class="card">
      <h2>CSS</h2>
      <p>Web page styling</p>
      <button>Learn More</button>
    </div>

  </div>

</body>
</html>
```

2. Complete Stylesheet (`style.css`)

```
body {
  background-color: #f5f5f5;
  font-family: Arial, sans-serif;
  margin: 0;
  padding: 20px;
}

h1 {
  text-align: center;
  color: purple;
  font-size: 32px;
}

.container {
  display: flex;
  justify-content: center;
  flex-wrap: wrap;
  gap: 20px;
  margin-top: 30px;
}

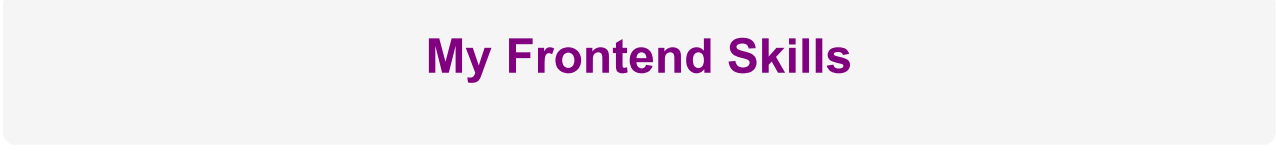
.card {
  width: 220px;
  background-color: white;
  padding: 20px;
  border: 2px solid black;
  border-radius: 10px;
  text-align: center;
}

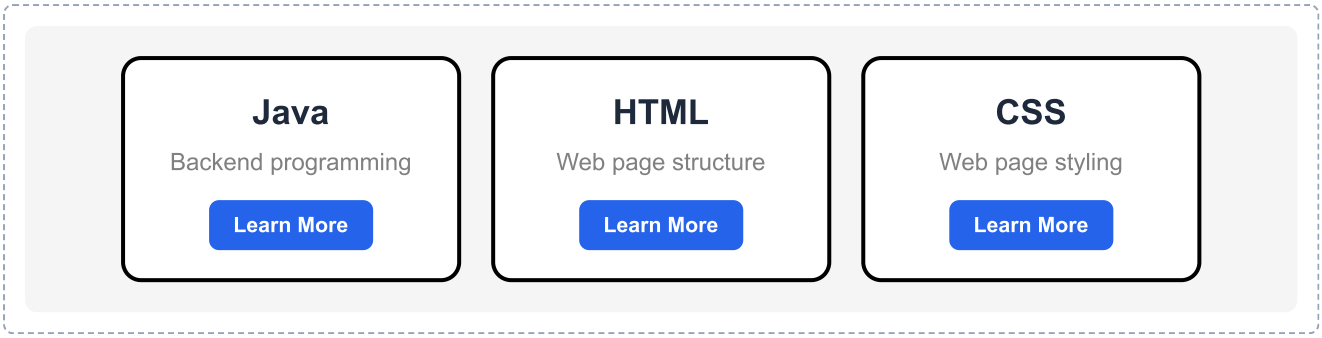
.card h2 {
  color: #0f172a;
  margin-top: 0;
}

.card p {
  color: gray;
  font-size: 14px;
}

.card button {
  background-color: rgb(37, 99, 235);
  color: white;
  border: none;
  border-radius: 5px;
  padding: 8px 16px;
  cursor: pointer;
  margin-top: 10px;
}
```

RENDERED BROWSER OUTPUT PREVIEW:





Chapter 14: Ten Critical Beginner CSS Mistakes & Fixes

Common Anti-Pattern	Erroneous Code	Corrected Code & Explanation
1. Styling element as class	`.body { margin: 0; }`	`body { margin: 0; }` HTML element selectors never use a leading dot (`.`).
2. Space inside color keyword	`background-color: light blue;`	`background-color: lightblue;` Spaces split tokens, invalidating the declaration.
3. Missing unit quantity	`padding: px;`	`padding: 10px;` Units require a numerical value before the unit suffix.
4. Missing semicolon	`color: purple font-size: 20px;`	`color: purple; font-size: 20px;` Every CSS declaration must end with a semicolon (`;`).
5. Wrong alignment property	`justify-content: center;` for vertical center in a row	`align-items: center;` Cross-axis (vertical in a row) is controlled by `align-items`.
6. Confusing alignment targets	Using `align-content` on single lines	Use `align-items` for items within a line; use `align-content` only for multiple flex lines.
7. Forgetting flex on parent	Adding `justify-content` directly to child	`display: flex;` must be applied to the **parent container**.
8. Unanchored absolute child	`position: absolute;` without relative parent	Add `position: relative;` to the parent card to establish a containing block.
9. Confusing margin & padding	Using margin when padding is needed	Padding lives *inside* the border; margin lives *outside*.
10. Missing dot on class selector	`btn { background: blue; }`	`.btn { background: blue; }` Class selectors must always begin with a period (`.`).

Chapter 15: 30 Technical Revision Questions

Part 1: Multiple Choice Questions (1–15)

1. What does CSS stand for? (A) Computer Style Sheets (B) Creative Style Sheets (C) Cascading Style Sheets (D) Colorful Style Sheets
2. Which symbol precedes a CSS class selector? (A) `#` (B) `.` (C) `&` (D) `*`
3. Which symbol precedes an ID selector? (A) `.` (B) `#` (C) `@` (D) `/`
4. Which part of the Box Model sits directly between content and border? (A) Margin (B) Outline (C) Padding (D) Gutter
5. In `padding: 10px 20px 30px 40px;`, what does 20px represent? (A) Top (B) Right (C) Bottom (D) Left
6. Under standard content-box sizing, if width is 200px, padding is 10px, and border is 2px, what is the total rendered width? (A) 200px (B) 212px (C) 224px (D) 230px
7. Which property changes text color? (A) `font-color` (B) `text-color` (C) `color` (D) `background`
8. Which display value flows side-by-side while respecting explicit width and height? (A) block (B) inline (C) inline-block (D) none
9. What is the default CSS positioning scheme for HTML elements? (A) relative (B) absolute (C) static (D) fixed
10. Which positioning scheme removes an element from normal flow and anchors it to the viewport? (A) relative (B) fixed (C) absolute (D) sticky
11. What happens when `position: relative; top: 10px;` is applied? (A) The element shifts down 10px and retains its original space (B) The element is removed from flow (C) The parent shrinks (D) The element pins to viewport
12. To anchor an absolutely positioned child to its parent, what position must the parent have? (A) static (B) non-static, typically relative (C) fixed only (D) absolute only

13. Which property aligns flex items along the main axis? (A) `align-items` (B) `justify-content` (C) `align-content` (D) `flex-flow`
14. What is the main axis when `flex-direction: row;` is declared? (A) Vertical (B) Horizontal (C) Diagonal (D) Circular
15. When does `align-content` take effect? (A) Always (B) Only on buttons (C) Only when `flex-wrap: wrap;` creates multiple lines (D) Only on tables

Part 2: Short Answer Conceptual Questions (16–25)

16. Explain why external CSS linked via a `<link>` tag is preferred over inline CSS.
17. Why can multiple elements share the same class name, but only one element should have a given ID?
18. Does margin change the internal clickable area of an HTML button? Explain why or why not.
19. What does the unitless value `line-height: 1.5;` mean relative to font size?
20. Why does the browser fail to parse `background-color: light blue;` ?
21. What does the CSS rule `border: none;` achieve when applied to a `<button>` ?
22. What is the semantic meaning of a `<div>` tag, and when should it be used?
23. Why are `width` and `height` properties ignored on standard `<span>` tags?
24. What is the difference between `justify-content: center;` and `align-items: center;` in a row layout?
25. How does `gap: 20px;` simplify spacing compared to applying margins to flex items?

Part 3: Practical CSS Analysis & Math Problems (26–30)

26. A container has width 300px, `display: flex; flex-wrap: wrap; gap: 10px;`. It contains boxes with width 140px and border 2px on each side. How many boxes fit per row?
27. Write the 2-value shorthand for an element that needs 20px vertical margin and 40px horizontal margin.
28. Spot and correct the bug: `<style> .btn { color: white background-color: blue; } </style>`.
29. A notification badge needs to sit in the top-right corner of a card. Write the necessary CSS position rules for both the card and the badge.
30. Explain why three 100px boxes with 20px gaps wrap inside a 310px flex container when `flex-wrap: wrap;` is enabled.

Chapter 15 Answer Key

Answers: Part 1 & Part 2 (1–20)

1: C (Cascading Style Sheets) | **2:** B (Period `.`) | **3:** B (Hash `#`) | **4:** C (Padding) | **5:** B (Right side, clockwise TRBL) | **6:** C (200 + 20 padding + 4 border = 224px) | **7:** C (`color`) | **8:** C (inline-block) | **9:** C (static) | **10:** B (fixed) | **11:** A (Shifts from natural position, reserves original space) | **12:** B (Non-static ancestor, usually relative) | **13:** B (`justify-content`) | **14:** B (Horizontal) | **15:** C (When multiple lines exist via wrapping).

16: External CSS separates concerns, allows styling multiple pages from a single file, and leverages browser caching.

17: Classes define reusable styles for many elements; IDs are unique DOM identifiers for specific elements.

18: No. Margin sits outside the border and is transparent; only padding expands the element's clickable area.

19: The line height is 1.5 times the current font size (e.g. 16px font * 1.5 = 24px line height).

20: Spaces separate values in CSS syntax; `lightblue` is a valid single token keyword.

Answers: Part 2 & Part 3 (21–30)

21: It removes default native browser button styling (the 3D beveled gray border).

22: `<div>` has zero semantic meaning; use it purely as a layout or styling grouping wrapper.

23: `<span>` is an inline element; inline elements flow with text and do not support block dimension properties.

24: `justify-content` centers horizontally along the row; `align-items` centers vertically within the row height.

25: `gap` applies spacing strictly between items without adding unwanted outer margin to the first or last elements.

26: Each box is 140px + 4px border = 144px. Two boxes + one 10px gap = 144 + 144 + 10 = 298px <= 300px. Exactly **2 boxes** fit.

27: `margin: 20px 40px;`

28: Missing semicolon after white: `.btn { color: white; background-color: blue; }`

29: Parent: `.card { position: relative; }` | Child: `.badge { position: absolute; top: 0; right: 0; }`

30: Three 100px boxes (300px) + two 20px gaps (40px) = 340px required. 340px > 310px, forcing the 3rd box to wrap.

Chapter 16: Quick Revision CSS Architecture Cheat Sheet

1. Syntax & Selectors

- **Rule:** `selector { property: value; }`
- **Element:** `h1 {}` (Targets all matching HTML tags)
- **Class:** `.card {}` (Reusable styling for multiple elements)
- **ID:** `#header {}` (Strictly unique single-element target)

2. Box Model Geometry Formula

```
Rendered Width = width + padding-left + padding-right + border-left + border-right
```

Margin sits outside the border and creates space between elements.

4. Display & Positioning Matrix

3. Margin & Padding Shorthand (Clockwise)

- `padding: 10px;` (All 4 sides)
- `padding: 10px 20px;` (Vertical, Horizontal)
- `padding: 10px 20px 30px;` (Top, Horizontal, Bottom)
- `padding: 10px 20px 30px 40px;` (Top, Right, Bottom, Left)

- `block` : New line, expands to full width, respects dimensions.
- `inline` : Flows with text, ignores width/height.
- `inline-block` : Flows side-by-side, respects width/height.
- `relative` : Offsets from self, reserves original layout space.
- `absolute` : Pulled from flow, anchored to nearest positioned parent.
- `fixed` : Pulled from flow, anchored permanently to viewport.
- `sticky` : Normal flow until scroll threshold, then locks in place.

5. Flexbox Quick Command Reference

- `display: flex;` : Turns parent into flex container.
- `flex-direction: row | column;` : Sets main axis orientation.
- `justify-content` : Aligns along main axis (`center` , `space-between`).
- `align-items` : Aligns along cross axis within each line.
- `gap: 20px;` : Clean gutter between items.
- `flex-wrap: wrap;` : Allows overflowing items to wrap onto new rows.
- `align-content` : Distributes multiple lines (requires wrapping).

Next Step in Your Roadmap

You have mastered **CSS Fundamentals, the Box Model, Positioning, and Flexbox** on Day 3. Moving into Days 4–5 according to your roadmap, you will deepen your Flexbox proficiency with real-world layouts (responsive navigation bars, centered login panels, and card grids) before moving into CSS Grid and mobile-first responsive media queries.